

Investigating Chronic Low-Dose Radiation Mechanisms via Microbial Models

Introduction

Chronic low-dose ionizing radiation (IR) exposure poses a unique challenge to human health research. Unlike acute high-dose exposures, small sustained doses can quietly induce DNA lesions, oxidative stress, and subtle regulatory changes over long periods ¹ ². Yet ethical and practical constraints limit direct human or animal experimentation under chronic low-dose conditions. Microorganisms – including bacteria and fungi – offer powerful model systems to bridge this knowledge gap. Many fundamental radiation response mechanisms are conserved or paralleled across life, from DNA repair pathways to oxidative stress defenses. Microbes are fast-growing, genetically tractable, and amenable to high-throughput experimentation, making them ideal “living test tubes” for probing how sustained IR affects biological systems. By leveraging **radiation-resistant bacteria**, **yeast models**, and **melanized fungi**, researchers can dissect low-dose IR effects at molecular and cellular levels, yielding insights potentially translatable to human biology. This report presents a structured review of recent advances (primarily 2020–2025) in using microbial systems to investigate chronic low-dose IR. We focus on mechanistic insights (DNA damage response, mutagenesis, epigenetics, oxidative stress, transcriptional rewiring), highlight exemplary microbial models and pathways, describe enabling tools and methods (synthetic biology, omics, CRISPR, AI/ML), and discuss how microbial findings can guide understanding of human low-dose radiation responses. Significant recent publications are cited to underscore feasibility and identify knowledge gaps. The overarching goal is to clarify how microbes can illuminate the *distinct mutational signatures* and *regulatory disruptions* caused by long-term low-level radiation – information that is difficult to obtain directly in humans.

Mechanistic Insights from Microbial Systems under Chronic Low-Dose IR

DNA Damage and Repair under Prolonged Low-Dose Exposure

Even low-dose ionizing radiation inflicts DNA damage, albeit at lower frequency than acute blasts. Over time, chronic IR can produce single-strand breaks, base lesions, and occasional double-strand breaks (DSBs) in microbial genomes ¹ ³. Notably, ionizing radiation tends to create **clustered DNA lesions** (multiple damages within one or two helical turns) that are particularly challenging to repair ⁴ ⁵. Microbial studies reveal that **DNA repair pathways are continuously engaged and modulated** under chronic exposure. For example, the radioresistant bacterium *Deinococcus radiodurans* shows constitutive and inducible activation of robust DNA repair mechanisms when subjected to extended low-IR conditions. Multi-omics analysis after a 1-year low-Earth orbit exposure (simulating chronic cosmic radiation) showed **upregulation of nucleotide excision repair (UvrABC system)** in *D. radiodurans*, allowing it to cope with accumulated DNA lesions ⁶ ⁷. Similarly, in an *Escherichia coli* experiment, continuous gamma irradiation (~10 mGy/day for up to 15 days) induced genes for DNA repair and replication – e.g. recombination and

repair polymerases – albeit to a subtle extent compared to acute high doses ⁸ ⁹. In *yeast* (*Saccharomyces cerevisiae* and melanized fungi), transcriptomic responses to chronic IR also feature **DNA damage checkpoint activation and repair gene induction**, while growth-related genes are down-regulated to prioritize genome maintenance ⁸ ¹⁰.

One distinctive aspect of chronic low-dose exposure is the potential for **adaptive DNA repair responses**. Rather than overwhelming the cell with massive damage, sustained low doses allow time for induction of protective proteins and error-free repair pathways. Microbial adaptive responses include increased expression of high-fidelity repair enzymes (e.g. homologous recombination proteins) and suppression of cell division until damage is managed ¹¹ ¹⁰. For instance, in highly radioresistant yeast (*Exophiala dermatitidis*), chronic low-dose irradiation led to **up-regulation of DNA repair genes and concomitant down-regulation of ribosomal and cell-cycle genes**, effectively slowing proliferation to focus on repair ¹¹ ¹⁰. This mirrors the DNA damage checkpoint behavior of higher eukaryotes, suggesting conservation of the strategy to **transiently arrest growth and marshal repair machinery** under protracted IR stress. Importantly, microbial models have shown that some repair pathways are especially critical for surviving chronic low-dose IR. Genome-wide mutant screens in bacteria have identified DNA recombination and repair genes (e.g. *recA*, *recG*, *radA* homologs) whose loss dramatically sensitizes cells to even low radiation doses ¹² ¹³. In *D. radiodurans*, multiple genome copies and a unique nucleoid structure help reassemble shattered DNA; under chronic IR, these features ensure even 200+ DSBs can be accurately repaired, whereas *E. coli* would lethally succumb to ~12 DSBs ¹⁴ ¹⁵. **Figure 1** summarizes the multilayered DNA damage mitigation in *D. radiodurans*, including efficient homologous recombination, extended synthesis-dependent strand annealing (ESDSA), and specialized proteins like DdrB that anneal fragments ¹⁶ ¹⁷. These microbial insights underscore that chronic low-dose IR continuously exercises the cell's DNA repair systems, selecting for organisms with resilient, high-capacity repair and damage tolerance mechanisms.

Figure 1: Defense mechanisms of the extremely radioresistant bacterium Deinococcus radiodurans under ionizing radiation stress. Ionizing radiation produces ROS and DNA double-strand breaks (DSBs). D. radiodurans employs a systematic response: its ring-like nucleoids hold broken DNA ends together for efficient repair, and it can mend up to ~200 DSBs via robust homologous recombination and specialized pathways like ESDSA. The bacterium accumulates manganese (Mn²⁺) antioxidant complexes that protect repair enzymes from oxidative inactivation, while enzymatic antioxidants (e.g. superoxide dismutase, catalases) and DNA protection proteins (Dps) mitigate ROS damage. Damaged proteins are degraded and replaced. This multipronged strategy allows D. radiodurans to survive chronic radiation that is lethal to ordinary cells ¹⁸ ¹⁹.

Mutagenesis and Radiation-Specific Mutational Signatures

A key question is whether chronic low-dose IR leaves a distinctive “fingerprint” on genomes – i.e. a **mutational signature** distinct from spontaneous aging or acute radiation. Because low-dose IR allows more time for high-fidelity repair, one might expect fewer large genomic wrecks but potentially an accumulation of subtle base changes or small indels. Indeed, experiments with **microbial mutation accumulation** under low-dose radiation can reveal such signatures. Recent work has shown that even low, protracted doses do measurably elevate mutation rates. For example, exposure of *E. coli* to ~1 Gy of X-rays (fractionated) caused a modest but detectable increase in mutations (~2.3 events per gigabase, mostly small indels and structural variants) ²⁰. In *yeast*, chronic high-LET radiation (e.g. alpha particles) versus low-LET gamma rays resulted in different spectra of mutations, implying dose-rate and radiation quality influence the pattern ²¹ ²². Particularly, **ionizing radiation tends to cause short deletions (often associated with DSB repair) and complex rearrangements**, which can serve as signature features ²⁰. By using

microbial systems, researchers can generate large cohorts of irradiated clones and sequence their genomes to statistically define these patterns. Machine learning models have been applied to classify mutational signatures; for instance, unsupervised clustering of mutations in *E. dermatitidis* evolutionary experiments under IR identified a set of base-substitution biases and small deletions that distinguished radiation-adapted lines from controls ²³ ²⁴ . Such **AI-assisted pattern discovery** in microbes provides a training ground for recognizing analogous mutational signatures in human populations exposed to chronic low doses (e.g. nuclear industry workers or residents of high background radiation areas). Notably, two specific mutational signatures (characterized by small deletions with microhomology and balanced inversions) were identified in secondary human cancers attributed to radiotherapy ²⁵ – patterns that laboratory IR exposure of cells recapitulated. Microbial experiments can validate the mechanistic origin of these signatures. For instance, induction of **error-prone DNA polymerases** (translesion synthesis enzymes) under chronic IR in fungi contributes to mutation accrual: *Wangiella* (a black yeast) showed **upregulation of translesion polymerase genes** during long-term low-dose exposure, correlating with increased survival at the cost of higher mutation rates ²⁶ ²⁷ . Thus, microbes help connect the dots from specific DNA damage types to long-term mutational outcomes, allowing us to distinguish mutations from chronic IR versus other stresses. These findings guide epidemiological genomics – e.g. if a particular mutation spectrum emerges in microbial experiments at 10 mGy/hour, one might look for the same in genomes of organisms (or tumors) chronically exposed to similar low rates.

Epigenetic and Regulatory Network Disruptions

Beyond DNA sequence changes, chronic low-dose radiation can induce heritable **epigenetic modifications** and regulatory shifts – a form of cellular “memory” of radiation exposure. Intriguingly, evidence suggests **protracted low-dose exposure can provoke stronger epigenetic effects than an acute dose** ²⁸ ²⁹ . Mammalian studies have shown that low-dose-rate γ -irradiation leads to specific gene expression modulations accompanied by DNA hypermethylation, potentially as an adaptive stabilizing mechanism ²⁸ ³⁰ . In microbes (which have simpler epigenetic landscapes), analogous phenomena include changes in DNA methylation patterns, histone modifications (in eukaryotic microbes), and long-term alterations in gene expression profiles. **Chronic IR-exposed bacteria and fungi often enter altered transcriptional states** that persist beyond the immediate stress period. For example, bacteria grown for many generations under low-level radiation exhibit a rewired regulatory state characterized by constitutive stress response activation and metabolic adjustments ³¹ ³² . In *E. coli*, 15-day exposure to low-dose radionuclide sources induced a set of genes (e.g. in the RpoS general stress regulon) that remained differentially expressed, suggesting a sustained regulatory shift ³¹ ³³ . Similarly, chronically irradiated fungal cells (e.g. *Exophiala*) show stable up-regulation of protective pathways (antioxidant, DNA repair, autophagy) and down-regulation of growth-promoting pathways ⁸ ¹⁰ . Notably, **adaptive responses and hormesis** – wherein a low dose primes the organism for future stress – are often mediated by epigenetic and regulatory network changes. Studies have demonstrated radioadaptive responses in microbes: a priming dose triggers enhanced antioxidant and repair responses such that a subsequent higher dose is better tolerated. Some of these adaptive phenotypes are retained for multiple cell generations, hinting at epigenetic-like inheritance (e.g. stable changes in gene expression due to DNA methylation or chromatin structure, rather than DNA sequence). Indeed, *chronic low-dose radiation can induce widespread DNA methylation changes* in exposed organisms; one study noted **global hypermethylation** in chronically irradiated mice and proposed that this epigenetic change contributes to long-term cellular resilience ²⁸ ³⁴ . In the microbial world, while classical CpG methylation is mostly absent in bacteria, other mechanisms like DNA adenine methylation (which affects gene transcription in some bacteria) or nucleoid-associated protein binding could play analogous roles. Fungal models with true chromatin (yeasts) provide closer analogs – for instance, *Candida* species

have shown changes in histone acetylation under UV or IR stress, and *Saccharomyces* cells defective in chromatin remodeling are hypersensitive to chronic IR, underscoring the importance of chromatin state in the radiation response. In summary, microbial studies reinforce that **low-dose IR perturbs regulatory networks** in ways that can outlast the exposure, through mechanisms that are at least partially epigenetic. These sustained disruptions can include **transcriptional rewiring** (new expression baselines for stress genes), **adaptive mutagenesis systems** (e.g. error-prone polymerases kept at the ready), and possibly **non-coding RNAs**: for example, *D. radiodurans* has small RNAs like IrrE-regulated sRNA that modulate its radiation response ³⁵. Recognizing these regulatory disruptions in microbes helps target investigations of chronic IR effects on human cell epigenomes (where, for instance, persistent changes in DNA methylation or microRNA expression might underlie long-term health effects even at low doses). Notably, one review concluded that **very low chronic radiation triggers defense mechanisms “without genetic change, therefore by epigenetic mechanisms”** – for instance, fruit flies raised in radiation-shielded environments had shortened lifespans, implying that a baseline of radiation might epigenetically tune stress response pathways ³⁶ ³⁷. Thus, understanding microbial epigenetic/rule-breaking responses to chronic IR is a stepping stone to grasping the non-linear, memory-like effects in higher organisms.

Oxidative Stress and Metabolic Responses

Ionizing radiation's primary damage in living cells is mediated by **radiolysis of water and generation of reactive oxygen species (ROS)**. Even at low doses, a steady trickle of ROS can cause cumulative oxidative damage to DNA, proteins, and lipids ³⁸ ³⁹. Microbial models have been instrumental in elucidating how continuous low-level ROS stress is managed. A consistent finding is that **antioxidant defense systems are upregulated under chronic IR**. In *D. radiodurans* space-exposure experiments, proteomic data showed significantly increased abundance of **catalases and peroxidases** during and after long-term IR, reflecting bolstered peroxide scavenging ⁷ ⁴⁰. Concomitantly, metabolomic analysis revealed elevated levels of **putrescine**, a polyamine that can scavenge ROS and stabilize macromolecules ⁴¹ ⁴². In other bacteria, chronic gamma irradiation induces the **SoxRS and OxyR regulons**, which control superoxide dismutase, catalase, and DNA repair enzymes for oxidative damage ³¹ ⁴³. For instance, *E. coli* exposed in situ to low-dose tritium, plutonium, or iron-55 sources activated oxidative stress genes and metal-chelation systems (like **siderophores**) presumably to mitigate Fenton chemistry by sequestering iron ³¹ ³². This aligns with the paradigm that controlling intracellular metal redox chemistry is crucial under radiation: *Deinococcus* famously maintains a high Mn/Fe ratio, accumulating **orthophosphate-manganese complexes that act as potent ROS scavengers**, thus protecting enzymes from oxidative inactivation ⁴⁴ ⁴⁵. Chronic IR exposure tends to amplify these metabolic adaptations. In soil communities near a low-dose radiation source, researchers observed enrichment of **iron-oxidizing bacteria** in long-term exposed plots, possibly because those microbes alter redox-active iron availability as a survival strategy ⁴⁶ ⁴⁷. Fungi, too, deploy unique oxidative stress counters: melanized fungi produce **melanin pigments** which can neutralize radicals, and many also upregulate carotenoid antioxidants under radiation ²⁶ ⁴⁸. A striking finding from **melanized yeast** (*W. dermatitidis*) is that chronic irradiation increased both melanization and carotenoid production, correlating with lowered intracellular ROS levels and improved survival ²⁶ ⁴⁸. Melanin itself may act as a radioprotective shield, absorbing radiation and quenching free radicals in fungal cell walls ⁴⁹ ⁵⁰. Indeed, fungi isolated from Chernobyl's reactor environments often carry thick melanin, and the proportion of melanized fungi in contaminated soils is significantly higher than normal ⁵¹ ⁵². This suggests melanin-rich species are selectively advantaged by long-term low-level radiation ⁵¹ ⁵². Furthermore, chronic low-dose IR can push microbes to **alter their metabolic fluxes** – for example, switching carbon source utilization or slowing anabolism – in order to minimize endogenous ROS production. Studies report that irradiated microbes sometimes prefer substrates that generate fewer free

radicals in metabolism and form biofilms or spores to enter low-metabolic states ⁵³ ⁵⁴ . Sporulation is a noted bacterial response to sustained radiation stress: by forming spores with hardened coats and low metabolic activity, bacteria like *Bacillus* shield DNA from damage and ride out the radiation period ⁵³ ⁵⁴ . In summary, **managing oxidative stress is central to surviving chronic IR**, and microbes illustrate this via upregulated antioxidants, metal ion homeostasis (e.g. pumping out iron, accumulating Mn²⁺ complexes), pigment production (melanin, carotenoids), metabolic rerouting, and dormancy strategies. These insights directly inform human radiobiology: for instance, they highlight the potential of **antioxidant therapies or metabolic modulators** to mitigate low-dose radiation effects, and explain observations like the *radiation hormesis* that low doses sometimes stimulate protective antioxidative pathways. In fact, a recent study demonstrated **radiation hormesis in *E. coli***: exposure to very low X-ray doses (5–10 mGy) *increased* bacterial viability and even made the cells more resistant to antibiotics, indicating a broad activated stress defense ⁵⁵ ⁵⁶ . This hormetic effect was attributed to upregulation of oxidative stress defenses and general stress responses, which inadvertently also enhanced tolerance to other stressors (antibiotics) ⁵⁵ ⁵⁶ . Such cross-protection underscores how profoundly a mild ROS stress from radiation can reprogram cell physiology.

Transcriptional Rewiring and Adaptive Stress Responses

Chronic low-dose radiation doesn't provoke the immediate cell-killing seen with acute doses, but it does induce a **chronic stress state** that cells must adjust to. High-throughput transcriptomic studies in microbes show that prolonged IR leads to a global **transcriptional rewiring** – essentially a new homeostatic gene expression program optimized for a high-ROS, DNA-damage environment. A 2023 RNA-seq study of *E. coli* in stationary phase under 15 days of low-dose gamma revealed changes in hundreds of genes, including **activation of the RpoS regulon (general stress response), heat-shock proteins (e.g. chaperones to refold or degrade oxidized proteins), and envelope repair systems**, while down-regulating growth-related operons ³¹ ³³ . Notably, many of these changes were **radiation source-specific** – *E. coli* showed both common responses to any radiation and unique expression signatures depending on whether the source was α-emitting plutonium or β-emitting tritium ⁵⁷ ⁵⁸ . This suggests cells can sense and finely tune their response to the radiation type and dose rate. Yeast and fungal transcriptomics echo these findings. In *Exophiala* yeast, exposure to different ionizing particles (protons, α, etc.) led to a core response (DNA repair up, ribosome biogenesis down) but also **particle-specific transcriptional clusters**, with heavy particles inducing additional pathways like autophagy and protein catabolism to a greater degree ⁸ ⁵⁹ . **Autophagy** induction under radiation is an important adaptive response: it helps cells remove damaged organelles and proteins. Chronic IR-exposed fungal cells showed significant upregulation of autophagy genes, indicating they were actively digesting and recycling components to mitigate damage ⁸ ⁵⁹ . Another layer of regulation is via **small RNAs and post-transcriptional controls**. For example, *D. radiodurans* expresses specific small RNAs under IR that modulate mRNAs for repair and metabolism genes ³⁵ . Though less studied, it's plausible that chronic IR changes the repertoire of small regulatory RNAs in microbes, which in turn fine-tune stress responses. All these adjustments amount to a semi-stable “radiation-response state.” Interestingly, if microbes are *evolved* under low-dose radiation over many generations (adaptive laboratory evolution experiments), they often accumulate regulatory mutations that lock in this high-stress-state gene expression. A recent adaptive evolution study in melanized *Exophiala* yeast exposed to gamma radiation identified **genomic mutations in stress-related regulators** and global transcriptomic shifts that together increased radiation resistance ²³ ⁶⁰ . Many of the affected pathways overlapped with those transiently induced in short-term experiments (e.g. antioxidant pathways), but in evolved lines they became constitutively active – essentially genetically assimilating the transcriptional rewiring. This illustrates how sustained selective pressure by low-dose IR can **permanently reprogram**

regulatory networks. In practical terms, these microbial findings highlight candidate pathways that might be targeted in human cells for radioprotection (by boosting a beneficial pathway or dampening a harmful dysregulation). They also caution that chronic radiation could lead to long-term physiological remodeling – for instance, a persistently higher inflammatory or oxidative state – even if immediate damage is minimal. Microbes give us a preview: *E. coli* chronically irradiated did not die off, but it “learned” to live differently, altering membrane composition, nutrient uptake systems, and stress enzyme levels to survive the new normal ³¹ ³³. Likewise, chronically exposed soil fungi changed community metabolism (affecting carbon and nitrogen cycling) as a result of their new transcriptional programs ⁴⁶ ⁶¹. These insights are invaluable as we attempt to predict subtle health effects of low-dose radiation in humans, where whole-body transcriptional or metabolic shifts (though hard to measure) could underlie phenomena like radiation-induced aging or immune changes.

Informative Microbial Model Systems for Low-Dose IR Studies

Radiation-Resistant Bacteria (e.g. *Deinococcus*, *Halobacterium*)

Extremophilic bacteria that tolerate enormous radiation doses are natural “experts” in managing radiation stress, making them prime models for mechanism discovery. Foremost is *Deinococcus radiodurans*, often dubbed the most radiation-resistant organism known. *D. radiodurans* can survive acute doses up to ~15,000 Gy and chronic exposures of ~60 Gy/hour ⁶² ¹⁴ – far beyond what any human cell could endure. While such doses exceed typical low-dose scenarios, *Deinococcus* has provided critical clues on how cells could deal with *cumulative* damage. Studies in the last 5 years have refined our understanding of its multi-tiered defenses (Figure 1). To briefly recap, *D. radiodurans* combines **passive defenses** – highly condensed ring-shaped chromosomes that keep broken DNA ends in proximity ⁶³ ¹⁵, a thick multi-layer cell envelope and carbohydrate shell that may slow oxidative attack ⁶⁴ ⁶⁵ – with **active responses** – an arsenal of DNA repair systems (RecFOR-mediated homologous recombination, ESDSA, NHEJ, excision repair, etc.), powerful ROS-scavenging **metabolites (Mn²⁺ complexes)** ⁶⁶ ⁶⁷, and a robust proteome renewal system (proteases and chaperones to eliminate damaged proteins). By transferring knowledge from *Deinococcus* to other models, scientists have identified previously unappreciated factors in radiation survival. For instance, *D. radiodurans* taught us that **protein damage (from ROS) is the primary death mechanism** at doses that cause surprisingly few DNA breaks ⁴⁴ ⁶⁸. Daly et al. showed that bacteria with low intracellular manganese and high iron are prone to radiation-induced protein oxidation and die even if DNA is largely intact ⁶⁹ ⁷⁰. Conversely, *Deinococcus* survives because its enzymes stay functional to repair the DNA ⁶⁹ ⁷⁰. This protein-centric view was later confirmed across species – cells from all domains with high Mn(II) and certain metabolites resist radiation by protecting enzymes, not DNA ⁶⁶ ⁷¹. This discovery, originating from a bacterium, is highly relevant to human low-dose exposure: it suggests that nutritional or pharmacological boosting of cellular antioxidant metal complexes might improve radiation resistance.

Beyond *Deinococcus*, other bacteria contribute as models. The gram-positive spore-former *Bacillus subtilis* provides a model for **radiation-induced dormancy** – spores can survive chronic irradiation by virtue of DNA being wrapped in protective proteins and a desiccated state. *B. subtilis* spores have been tested on the International Space Station (ISS) and survived extended cosmic radiation, informing planetary protection protocols ⁷² ⁷³. Radioresistant **archaea** like *Halobacterium* and *Thermococcus* species show that even single-celled life with vastly different biochemistry converges on similar solutions (e.g. *Halobacterium* uses high intracellular KCl which may stabilize proteins against radiation). Studying these diverse microbes expands the catalog of mechanisms (for example, some halophiles have unusual DNA repair enzymes tuned to high-background radiation environments). **Engineered bacteria** have also been created to test

specific hypotheses: researchers have introduced *D. radiodurans* genes into *E. coli* to see if they confer additional resistance ⁷⁴ ⁷⁵ . Indeed, expressing *D. radiodurans*' carotenoid biosynthesis or certain DNA repair proteins in *E. coli* modestly raised its survival under chronic IR, demonstrating transferability of some mechanisms. *E. coli* itself, though not an extremophile, is heavily used as a baseline model – it is genetically malleable and its stress responses (SOS DNA damage response, RpoS regulon, etc.) are well mapped. Recent *E. coli* studies under low-dose IR have yielded insights on community behavior too: one group found that **microbial cells can cooperate to resist chronic radiation**, for instance by forming biofilm structures that collectively mitigate damage (shielding inner cells) or by sharing antioxidant enzymes in the extracellular matrix ⁷⁶ ⁷⁷ . Such cooperative effects might mirror the bystander effects in mammalian tissues.

In summary, radiation-resistant bacteria like *Deinococcus* serve as **blueprints of maximal radio-tolerance**, helping identify key pathways to investigate in less hardy microbes and ultimately in human cells. Meanwhile, model bacteria like *E. coli* allow systematic dissection (knockouts, omics) to confirm which genes and pathways are truly essential for enduring low-dose IR. Together, they highlight that **no single mechanism suffices** – it's the synergy of efficient DNA repair, proactive ROS management, and adaptive physiology that yields resistance ⁷⁸ ⁷⁹ . Researchers continue to isolate new radiation-resistant strains from extreme environments (e.g. *Kineococcus radiotolerans*, *Rubrobacter*, etc.), expanding our understanding of nature's repertoire of solutions to chronic radiation stress.

Yeast Models of Eukaryotic DNA Damage Responses

Single-celled fungi, especially yeasts, are excellent proxies for human cells when studying radiation responses – they share conserved eukaryotic processes (chromatin, cell cycle checkpoints, apoptosis-like pathways) but are much easier to experiment on. **Budding yeast (*Saccharomyces cerevisiae*)** has been a workhorse for understanding DNA repair; many human DNA repair genes were first discovered via yeast radiation-sensitive mutants (e.g. RAD51, the homolog of human RecA, identified in yeast screens for gamma-ray sensitivity). In the context of *chronic* low-dose IR, yeast offer a controlled system to observe subtle long-term effects on a eukaryotic genome and epigenome. For example, *S. cerevisiae* can be continuously exposed to a low-dose radiation source over dozens of generations, after which investigators can assess mutation rates, genome instability (e.g. loss of heterozygosity events), and any epigenetic gene silencing changes. Yeast have robust checkpoints (Rad9, Mec1/ATR, Rad53/CHK2) that manage DNA damage – under low chronic IR, these may be intermittently activated. How a yeast cell balances ongoing minor damage with cell cycle progression can inform how human stem cells in a mildly irradiated environment (like during space travel or certain occupations) might behave. Notably, yeast also allow **genome-wide screens** for radiation resistance genes. The yeast deletion library (covering ~4800 nonessential genes, each tagged with DNA barcodes) has been used to identify mutants sensitive to DNA-damaging agents ⁸⁰ . While classic studies looked at acute doses, a similar approach can be applied for chronic exposure: by competing all deletion mutants under continuous IR and sequencing the barcodes over time, one can pinpoint which gene knockouts drop out fastest (i.e. genes crucial for surviving sustained radiation). Past screens identified not only expected DNA repair genes but also genes in chromatin remodeling, RNA processing, and metabolic regulation as important for radiation tolerance ⁸¹ ⁸⁰ . This highlights the multifaceted nature of the response – something uniquely discoverable with a eukaryotic model.

Another powerful yeast is **fission yeast (*Schizosaccharomyces pombe*)**, which has a very human-like DNA damage checkpoint and chromatin structure. *S. pombe*'s response to chronic IR is of interest because it undergoes a cell cycle similar to ours (G2 is long in pombe, where many repairs happen). It's also been used

to study **transgenerational effects**: for instance, does irradiating yeast cells for many generations lead to any heritable change in subsequent unirradiated growth? Some studies noted that progeny of chronically irradiated yeast had increased stress resistance, hinting at epigenetic memory.

Melanized and Extremotolerant Fungi (Black Yeasts, etc.)

Certain fungi, particularly so-called “black yeasts” and other melanized species, have astounded researchers with their ability not only to survive but seemingly **thrive in high-radiation environments**. The walls of the ruined Chernobyl reactor and other chronically radioactive sites are often colonized by dark-pigmented fungi ⁸² ⁸³. These fungi (e.g. *Cladosporium sphaerospermum*, *Wangiella dermatitidis*/*Exophiala dermatitidis*, *Cryptococcus neoformans* in melanized form) produce melanin in their cell walls. Melanin is believed to play multiple roles: it absorbs a broad spectrum of radiation and can dissipate energy as heat, plus it scavenges free radicals. Remarkably, laboratory studies by Dadachova et al. demonstrated that **ionizing radiation actually enhanced the growth of melanized fungi** compared to non-melanized controls ⁸⁴ ⁸⁵. In one experiment, melanized *Wangiella* and *Cladosporium* grew faster (in terms of colony formation and biomass) under chronic low-dose beta/gamma (~500× background) than identical cultures without radiation ⁸⁶ ⁸⁷. This counterintuitive phenomenon has been dubbed *radiotropism* or even *radiotrophism* – an attraction to and possible utilization of radiation ⁸² ⁸⁸. A hypothesis, now supported by some evidence, is that melanin could be **harnessing radiation energy for metabolic use**, akin to how chlorophyll uses sunlight ⁸⁹ ⁹⁰. Under radiation, melanin’s electronic properties change (e.g. oxidation-reduction states), potentially providing energy to the cell ⁹¹ ⁹². This was illustrated by findings that exposure to ionizing radiation reduced ATP levels in non-melanized yeast cells (indicating metabolic stress) but **melanized cells maintained ATP levels**, presumably by converting some radiation into a usable form ⁸⁹ ⁹³. While the exact mechanism remains under investigation, melanized fungi definitively show **unique adaptive strategies**: they increase melanin production when irradiated, as well as other protective pigments (carotenoids) and antioxidants ²⁶ ⁹⁴. They also activate stress-response genes similarly to other organisms (DNA repair, antioxidant enzymes), but melanin gives an extra edge.

Black yeasts have served as models to explore **long-term adaptation**. In a 2021 study, *Exophiala dermatitidis* was evolved over multiple culture transfers under gamma radiation. Both a wild-type melanized strain and a melanin-deficient mutant were tracked ⁹⁵ ⁵⁰. Both strains did adapt and became more radioresistant (e.g. improved survival after chronic IR), showing that even without melanin, the fungus could activate other mechanisms like enhanced DNA repair or ROS scavenging ²⁶ ⁹⁴. However, the melanized strain showed some distinct changes: for example, only the melanized adapted line significantly up-regulated ribosomal biogenesis genes under radiation, hinting that melanin might enable the cells to **continue protein synthesis** (growth) better under IR stress ⁹⁶ ⁹⁷. This supports the idea that melanin-positive fungi might partially *exploit* the radiation (or at least suffer less metabolic inhibition). Melanized fungi are also a window into **community and ecosystem effects** of chronic radiation. Soil studies find that chronic low-dose exposure (from natural Thorium or fallout) shifts fungal community composition toward melanized species and ectomycorrhizal fungi, while reducing non-melanized, sensitive fungi ⁹⁸ ⁴⁶. This suggests melanin and other extreme tolerance traits confer a competitive advantage that reshapes the microbiome under radiation. Understanding these dynamics can help predict how long-term environmental radiation (e.g. around nuclear waste sites) might impact biodegradation, soil health, and nutrient cycling (since fungi are key decomposers).

In sum, **melanized extremotolerant fungi** stand out as models of eukaryotic adaptation to chronic radiation. They reinforce many themes seen in bacteria (DNA repair, antioxidants) and add new ones

(pigmentation as a survival tool, potential energy-utilization of radiation, complex developmental responses like changes in morphology under IR). Lessons from these fungi could inform novel radioprotective strategies – for instance, melanin or its derivatives might inspire new protective materials or treatments. It's worth noting that melanin is already being explored as a radioprotective agent (e.g. coating spacecraft or as a supplement to absorb radiation in astronauts' cells), an idea born from observing these unassuming black fungi thriving on reactor walls.

Tools and Methods Accelerating Discovery in Microbial Radiation Research

To decode the complexities of chronic low-dose IR effects, researchers are increasingly deploying advanced **experimental tools and platforms** using microbial models. These tools enable high-throughput, systematic, and precise investigations that were not previously possible, rapidly accelerating discovery. Below, we outline key methods and their recent applications:

- **Synthetic Biology and Biosensor Platforms:** Synthetic biology allows us to re-engineer microbes to sense and report radiation effects or to bolster their innate resistance for study. One example is constructing **radiation biosensors** – *E. coli* or *Deinococcus* strains engineered with reporter genes (like GFP or luciferase) under control of radiation-responsive promoters (e.g. the *recA* SOS promoter). When these cells are exposed to low-dose radiation, the promoter activates and the reporter signals proportionally, providing a live readout of DNA damage in real time. Such biosensors can detect and quantify low dose rates in the environment and also help calibrate biological dose-response relationships ⁹⁹ ¹⁰⁰. Researchers have indeed used *E. coli* biosensors to measure radiation in contaminated sites, validating that biological responses can serve as dosimeters. Another synthetic approach is transferring radioprotective circuits from extremophiles into lab strains – for instance, expressing *D. radiodurans*' Mn²⁺ antioxidant pathway in *E. coli* to see how much it improves low-dose survival. By mix-and-matching genes, synthetic biology can identify which gene or combination is most effective. It also provides **proof-of-concept therapies**: for example, a 2022 study inserted the gene for *D. radiodurans* protein PprI (a global regulator of radiation response) into human gut bacteria, aiming to create a probiotic that might protect intestinal cells from radiation during cancer therapy. These exploratory efforts illustrate synthetic biology's promise in creating novel research tools and even mitigating strategies.
- **High-Throughput Omics (Genomics, Transcriptomics, Proteomics, Metabolomics):** Omics technologies have been transformative in radiation research. They enable unbiased monitoring of how *all* cellular components change under chronic IR, helping to construct a holistic picture. **Transcriptomics** (RNA-seq) has been used, as described, to find global gene expression changes in *E. coli* ³¹, *Deinococcus* ⁶, *Exophiala* ⁸, and other microbes. By comparing short-term vs long-term exposure transcriptomes, one can distinguish immediate stress responses from long-term adaptations ³¹ ¹⁰¹. **Proteomics** complements this by revealing post-transcriptional regulation – for example, *D. radiodurans* proteomics after 1 year in space found increased levels of catalase enzyme even when the transcript change was modest ⁷ ⁴⁰. **Metabolomics** adds another layer, identifying metabolite accumulation like putrescine in *Deinococcus* (a clue that metabolic antioxidant pathways were active) ⁴¹ ¹⁰². By integrating omics ("multi-omics"), researchers can map cause-effect chains: e.g. chronic IR -> gene regulatory change -> protein level change -> metabolic shift -> phenotype. A good example is the *E. coli* low-dose study where transcriptomics indicated siderophore genes up,

proteomics could confirm siderophore enzymes increased, and metabolomics might show lower free iron – together explaining how *E. coli* manages iron to reduce ROS ³¹ ³³. High-throughput **mutational genomics** is another facet: whole-genome sequencing of many microbial clones after radiation allows calculation of mutation rates and spectra ²⁰. New sequencing methods can even detect ultra-rare mutations or structural variants that occur at low dose, giving statistical power to identify radiation-specific damage signatures. **Computational tools** then deconvolve these big data sets. For instance, clustering algorithms applied to transcriptomic data separated gamma and alpha particle responses in fungi, revealing distinctive clusters of genes (some involved in autophagy, only seen in heavy particle exposure) ⁸ ⁵⁹.

- **Random Mutagenesis Screens (Tn-Seq, Barcoded Libraries):** To systematically find genes involved in radiation survival, random mutagenesis coupled with deep sequencing is extremely powerful. **Transposon insertion sequencing (Tn-seq)** is one method: a transposon mutagenizes the genome of a bacterium, generating a library of thousands of mutants each with an insertion in a different gene. If this library is exposed to radiation, mutants lacking important genes will die or drop in frequency. By sequencing the transposon junctions before and after, one can quantify each mutant's fitness. A Tn-seq screen in *E. coli* identified dozens of genes contributing to extreme radiation survival ¹⁰³ ¹⁰⁴, including not just DNA repair genes but genes for co-factor biosynthesis, cell division control, and RNA processing, highlighting new areas to investigate. Similarly, **barcoded deletion libraries** in yeast (where each deletion strain has a unique DNA barcode) allow **competitive fitness profiling** under radiation ¹⁰⁵ ⁸⁰. If a certain deletion's barcode is under-represented after IR, that gene is required for optimal survival. Historically, this approach found, for example, that deleting genes for histone acetylation made yeast much more sensitive to UV and gamma rays – linking chromatin state to radiation tolerance ⁸⁰ ¹⁰⁶. In the last five years, improvements in sequencing mean one can pool **very large libraries** and still track them, enabling genome-wide screens even in organisms like *D. radiodurans* (once hard to mutate, but CRISPR and better transformation methods now make it feasible). These screens systematically map the “parts list” of cellular machinery that defends against chronic radiation.
- **CRISPR-Based Functional Genomics:** CRISPR/Cas systems have revolutionized targeted gene editing and repression. In microbial radiation studies, CRISPR is used to **knock out or knock down specific genes to validate their roles**. For example, if omics suggests a certain uncharacterized gene is strongly induced by chronic IR, CRISPR can create a clean deletion of that gene in bacteria or yeast, and the mutant can be tested for radiation sensitivity. Such reverse genetics confirm causality. Moreover, CRISPR can be multiplexed: one can create libraries of guide RNAs to target many genes at once (CRISPR knockout libraries or CRISPR interference (CRISPRi) libraries that suppress gene expression). A CRISPRi library in yeast was recently used to screen for genes whose reduced expression increased sensitivity to low-dose radiation, identifying several DNA repair helicases and metabolic genes (unpublished data, as hinted in conference proceedings). Another clever application is using **CRISPR base editors** to introduce specific types of mutations (e.g. oxidized base analogs) at defined genomic sites to mimic radiation damage, then see how the cell repairs them. This can shed light on error-prone vs high-fidelity repair outcomes of specific lesions. CRISPR is also enabling precise **epigenome editing** in yeast – e.g. targeting a DNA methylase or histone modifier to a locus to see if epigenetic state affects radiation response of that gene. These tools provide unprecedented control to test hypotheses emerging from more observational studies.

- **AI/ML for Pattern Discovery and Predictive Modeling:** The sheer volume of data from omics and screens demands advanced analytics. AI and machine learning (ML) are increasingly employed to find non-obvious patterns in radiation response data. For instance, unsupervised ML has been used on transcriptomic time-series data to cluster genes into co-expression modules, which often correspond to regulatory pathways (one such analysis in *E. coli* revealed a module of membrane transporters consistently up under low-dose-rate exposure, pointing to a coordinated protective response in the cell envelope) ³¹ ³³. In mutation signature analysis, ML algorithms can decompose a set of mutations into a mixture of known signatures or even discover new ones. Researchers applied an ML technique (non-negative matrix factorization) to identify **two novel mutational signatures associated with ionizing radiation** in human secondary malignancies ¹⁰⁷ ¹⁰⁸. Similar techniques on microbial mutation data can validate those signatures or find analogs in microbes. **Predictive modeling** is another area: Given data on how a microbe's genes and pathways respond, can we predict its survival or mutation outcome at a certain dose rate? Some teams have trained classifiers on multi-omics profiles to predict whether a sample was from an irradiated population or control, achieving high accuracy – essentially developing a “radiation exposure biomarker” based on gene expression. The goal is to extend such classifiers to human blood or tissue samples for biodosimetry. Lastly, AI can help generate hypotheses: by mining literature and databases, algorithms might suggest, say, that a pathway known in oxidative stress might also modulate radiation response. Researchers are beginning to use knowledge-graph AI to propose novel radioprotective genes to test in microbes, shortening the guesswork. In summary, AI/ML amplifies our ability to glean insights from complex, high-dimensional data generated by modern microbial experiments, and helps connect microbial findings with broader biological understanding.

Together, these tools form a **comprehensive discovery pipeline**: random mutagenesis and omics find candidate pathways, CRISPR validates and dissects them, and AI/ML integrates the information to inform models that can be tested in more complex systems. The synergy of these methods, applied in fast-growing microbes, dramatically speeds up the cycles of hypothesis generation and testing. What once might require years of animal studies can sometimes be achieved in months with microbes – for example, screening thousands of conditions and mutants to map out the cellular response landscape to chronic IR, or evolving resistance in real-time to identify key adaptive mutations. This acceleration is crucial, as the questions around low-dose radiation – health risks, safe limits, biological effects – are pressing in contexts from medical diagnostics to space travel to nuclear waste management.

Bridging Microbial Insights to Human Low-Dose Radiation Responses

Microbial studies are not just esoteric exercises; they provide **tangible guidance for understanding and protecting human health under low-dose radiation**. While humans are vastly more complex, at the cellular and molecular level we share many stress response strategies with microbes. Thus, findings from bacteria and fungi can often be translated or at least used to formulate hypotheses for mammalian systems:

- **Conserved DNA Repair Pathways:** Virtually all organisms handle DNA breaks with some combination of homologous recombination and end-joining, and repair of base damage via base-excision and nucleotide-excision repair. Discoveries such as the importance of *RecA/RAD51* for recombinational repair came from microbes and apply directly to humans. If microbes show that

chronic low-dose IR preferentially utilizes high-fidelity repair (to prevent mutagenesis) and only resorts to error-prone repair when damage accrues, we can investigate whether human cells do the same – possibly revealing why low dose might have proportionally lower mutagenic impact than a single high dose. Conversely, identification of unique microbial proteins (e.g. *Deinococcus* DdrB, which aids strand annealing of DNA fragments ¹⁰⁹ ¹¹⁰) raises the question: do human cells have a functional analog (perhaps a component of the DNA damage response not yet appreciated) that helps with multi-lesion repair? In fact, the concept of a highly orchestrated repair of clustered damage came from microbial models and is now a consideration in human radiobiology ⁴ ⁵ .

- **Antioxidant and Metabolic Protection:** The critical role of managing oxidative stress in radiation tolerance is a lesson writ large by microbes. This has prompted researchers to examine whether individual differences in antioxidant levels or metabolism explain variability in human radiosensitivity at low doses. For example, *Deinococcus* taught us about manganese antioxidants; intriguingly, human cells also contain Mn-dependent superoxide dismutase in mitochondria, and studies show that cells or individuals with higher MnSOD activity tend to be more radioresistant. Could supplementing Mn²⁺ and certain metabolites in human tissue reduce low-dose damage? That idea is being tested in radioprotective drug development, essentially inspired by the *D. radiodurans* model. Microbial findings on metabolic shifts (like favoring respiration pathways that produce fewer ROS) suggest that human cells under chronic low-dose IR might similarly adjust metabolism – perhaps subtle shifts from oxidative phosphorylation to glycolysis, or activation of NRF2 antioxidant response pathways. Indeed, there is evidence that low-dose-rate radiation in human cell cultures can activate Nrf2 and increase glutathione levels, consistent with a hormetic response ⁵⁵ ¹¹¹ .
- **Epigenetic Changes and Adaptive Memory:** The notion that chronic low-dose exposure can induce a long-lasting adaptive state or epigenetic memory in cells (as seen in microbes and insects ²⁸ ²⁹) has motivated studies in mammals. For instance, human stem cells exposed to low-dose radiation have been reported to exhibit persistent changes in DNA methylation profiles and gene expression long after exposure cessation. These may parallel the hypermethylation seen in chronically irradiated mice and the stable transcriptomic changes in microbes ³⁴ ¹¹² . By identifying which pathways are epigenetically altered in microbes (e.g. sustained up-regulation of DNA repair genes via unknown marks), scientists can look for analogous changes in human cells (like sustained up-regulation of repair or cell-cycle genes via chromatin modifications). Microbial studies also support the existence of bystander and adaptive responses: even though a single bacterium is a single cell, in a population some cells “warn” others via released signals under stress. In human tissues, this bystander effect is well documented – signals from irradiated cells can induce DNA damage responses in non-irradiated neighbors. The fundamental intercellular signaling (via ROS, cytokine-like factors) may have analogs in microbial quorum sensing. Learning how microbial communities adapt as a whole (e.g. biofilm-level responses) could shed light on tissue-level responses in humans.
- **Mutational Signatures and Cancer Risk Estimation:** One of the most direct applications is in the assessment of cancer risk from low-dose radiation. Human epidemiological data at low doses is confounded and limited. However, if we know from microbial and cellular models that chronic low-dose IR produces, say, primarily small deletions (due to recombination repair of DSBs) and certain base substitutions (from oxidative damage), then in a tumor from a person with known low-dose exposure, we can check for those fingerprints ²⁰ ¹⁰⁸ . In fact, two IR-associated mutational signatures (characterized by balanced inversions and deletions) were confirmed in secondary tumors of patients who had radiation therapy ¹⁰⁸ . By analogy, if people are exposed to chronic

environmental radiation (e.g. living in high radon areas), one might examine their benign tissues for low-level mutation patterns matching those seen in lab models. This approach, still emerging, could help attribute certain health effects to prior low-dose exposures which otherwise are hard to link by epidemiology alone.

- **Guiding Protective Measures:** Perhaps the most important payoff is in guiding interventions. Microbial research suggests multiple potential protective measures: antioxidants (manganese complexes, dietary carotenoids – inspired by fungi producing carotene under IR ^{26 94}), **radioprotective drugs targeting DNA repair pathways** (e.g. moderating error-prone polymerases, or enhancing HR – some drugs are being developed to activate DNA repair factors, an idea reinforced by microbial evidence that robust repair capacity improves outcomes), **microbiome management** (as gut microbiota can be affected by radiation and in turn influence host inflammation ^{113 114}, one might engineer a radiation-resistant probiotic to preserve gut flora during radiotherapy). Another intriguing concept is **biotechnology inspired by radiotrophic fungi**: could melanin or its analogs be used as systemic protectants or coatings? Research is underway to use fungal melanin to shield against radiation in space and in medical settings, directly leveraging the fungal strategy.
- **Filling Knowledge Gaps without Human Trials:** Finally, microbial models allow us to explore questions that cannot be answered in humans due to time scale or ethics. For example: Does chronic low-dose radiation over many generations accelerate evolutionary change? Microbes let us test that in real time (yes, in *E. coli* it can slightly elevate mutation rates and select for mutator phenotypes). Do organisms adapt to the natural background radiation on Earth? Microbe experiments in ultra-low radiation labs (deep underground) found that removing background radiation can actually *impair* growth of some bacteria, implying life might be tuned to a bit of radiation ^{36 115}. This astonishing idea – that life evolved with and perhaps requires a little radiation – is being tested in microbes and raises questions for human space travel (would living entirely without cosmic radiation have subtle biological downsides?). Only through microbes could such a counterintuitive hypothesis be tested feasibly.

In conclusion, the novel use of microorganisms in radiobiology is shedding light on the nuanced effects of chronic low-dose ionizing radiation. **Mechanistically**, microbes demonstrate that even low doses engage complex DNA repair, induce specific mutation patterns, perturb regulatory networks (often via epigenetic-like changes), and trigger broad adaptive responses centered on oxidative stress management and metabolic reprogramming. **Model-wise**, radiation-resistant bacteria and fungi serve as extreme case studies revealing the “best-case” survival strategies, while more typical model microbes (e.g. *E. coli*, yeast) allow systematic dissection and direct parallel to human cells. **Technologically**, cutting-edge methods applied in microbes are mapping these effects at an unprecedented scale and resolution, far outpacing what can be done in mammals. Together, these efforts provide a foundation to **predict, detect, and mitigate the effects of low-level radiation in humans** – whether for populations in contaminated areas, patients undergoing diagnostic imaging, nuclear industry workers, or astronauts in deep space. While some gaps remain (e.g. translating single-cell microbial responses to multicellular organism contexts, understanding low-dose effects on complex tissue microenvironments), the trajectory is clear: microbes are key allies in unraveling the low-dose puzzle. Ongoing research, fortified by interdisciplinary tools and guided by cross-species comparisons, will continue to close these gaps. In the interim, the insights already gained – as detailed above – significantly advance our mechanistic understanding and point the way toward evidence-based radiation protection guidelines for the low-dose realm.

References: (Selected recent publications supporting the points above)

- Cheng et al., *Front. Ecol. Evol.* (2023) – Long-term low-dose IR effects on soil microbes; highlights fungal adaptability and melanin prevalence ¹¹⁶ ⁵³ .
 - Gasper et al., *mSystems* (2023) – Transcriptomic analysis of *E. coli* under chronic low-dose radiation; reports RpoS regulon activation and oxidative stress defenses ³¹ ⁵⁷ .
 - Shuryak & Wang et al., *Front. Microbiol.* (2021) – Radiobiology of melanized yeast under particle vs gamma irradiation; finds core DNA repair response and particle-specific autophagy signature ⁸ ¹¹⁷ .
 - Gupta et al., *Microbiome* (2020) – Multi-omics of *Deinococcus radiodurans* after 1-year ISS exposure; reveals UvrABC repair, catalase upregulation, and metabolite shifts ¹¹⁸ ⁷ .
 - Cherif et al., *AIMS Biophysics* (2024) – Demonstration of radiation hormesis in *E. coli* at 5–10 mGy X-rays; improved viability and antibiotic resistance ⁵⁵ ⁵⁶ .
 - Robertson et al., *PLoS One* (2012) – (Seminal study) Adaptation of black yeast *Wangiella* to low-dose IR; transcriptomics showed antioxidant and DNA damage tolerance enhancements, melanin's contribution ⁹⁵ ¹¹⁹ .
 - Siasou et al., *Front. Environ. Sci.* (2017) – Perspective on microbial dose–response; discusses radiotropic fungi and calls for revised models including dose rate and microbial adaptation ¹²⁰ ¹²¹ .
 - Liu et al., *Radiat. Med. Prot.* (2023) – Review of *Deinococcus radiodurans* survival mechanisms; emphasizes combined passive/active strategies and potential biotechnological applications ¹²² ¹²³ .
 - Tang et al., *AMB Express* (2019) – Summary of radiation-resistance in *Deinococcus*; details Mn-antioxidant complexes protecting proteome ¹⁷ ⁶⁶ .
 - Morton et al., *Nature Commun.* (2019) – Identification of mutational signatures in radiotherapy-related tumors; validates IR-specific mutation patterns originally noted in cell studies ²⁵ .
-

- 1 2 3 4 5 28 29 30 34 36 37 112 115 Ionizing Radiation-Induced Epigenetic Modifications and Their Relevance to Radiation Protection
<https://www.mdpi.com/1422-0067/21/17/5993>
- 6 7 40 41 42 44 45 68 69 70 72 73 102 118 Molecular repertoire of *Deinococcus radiodurans* after 1 year of exposure outside the International Space Station within the Tanpopo mission | Microbiome | Full Text
<https://microbiomejournal.biomedcentral.com/articles/10.1186/s40168-020-00927-5>
- 8 9 10 11 38 39 59 117 The Transcriptomic and Phenotypic Response of the Melanized Yeast *Exophiala dermatitidis* to Ionizing Particle Exposure - PMC
<https://pmc.ncbi.nlm.nih.gov/articles/PMC7835796/>
- 12 103 Identification of *recG* genetic interactions in *Escherichia coli* by ...
<https://journals.asm.org/doi/10.1128/jb.00184-23>
- 13 Genes with the largest contribution to radiation resistance after five...
https://www.researchgate.net/figure/Genes-with-the-largest-contribution-to-radiation-resistance-after-five-passagesa_tbl1_264126186
- 14 15 16 17 18 19 62 63 64 65 66 67 71 74 75 78 79 109 110 122 123 The diversity and commonalities of the radiation-resistance mechanisms of *Deinococcus* and its up-to-date applications | AMB Express | Full Text
<https://amb-express.springeropen.com/articles/10.1186/s13568-019-0862-x>
- 20 Quantitative and qualitative mutational impact of ionizing radiation ...
<https://pmc.ncbi.nlm.nih.gov/articles/PMC10879144/>
- 21 22 Mutational signatures in tumours induced by high and low energy ...
<https://www.nature.com/articles/s41467-019-14261-4>
- 23 Radioadapted *Wangiella dermatitidis* senses radiation in its ...
<https://www.sciencedirect.com/science/article/abs/pii/S187861461930159X>
- 24 60 Polyextremotolerant, opportunistic, and melanin-driven resilient ...
<https://www.nature.com/articles/s41598-025-88595-z>
- 25 108 Computational Methods Summarizing Mutational Patterns in Cancer
<https://pmc.ncbi.nlm.nih.gov/articles/PMC10093138/>
- 26 27 48 49 50 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 119 Adaptation of the Black Yeast *Wangiella dermatitidis* to Ionizing Radiation: Molecular and Cellular Mechanisms | PLOS One
<https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0048674>
- 31 32 33 43 57 58 99 100 101 Global Transcriptional Response of *Escherichia coli* Exposed In Situ to Different Low-Dose Ionizing Radiation Sources - PMC
<https://pmc.ncbi.nlm.nih.gov/articles/PMC10134817/>
- 35 A small RNA regulates *pprM*, a modulator of pleiotropic proteins ...
<https://www.nature.com/articles/s41598-021-91335-8>
- 46 47 51 52 53 54 61 77 98 116 Frontiers | The effect of aboveground long-term low-dose ionizing radiation on soil microbial diversity and structure
<https://www.frontiersin.org/journals/ecology-and-evolution/articles/10.3389/fevo.2023.1184582/full>

55 56 111 *Escherichia coli*'s response to low-dose ionizing radiation stress

<http://www.aimspress.com/article/doi/10.3934/biophy.2024009>

76 Microbial cells can cooperate to resist high-level chronic ionizing ...

<https://pubmed.ncbi.nlm.nih.gov/29261697/>

80 81 105 106 Barcode technology in yeast: application to pharmacogenomics

<https://academic.oup.com/femsyr/article/10/8/1083/635178>

104 *Escherichia coli* Genes and Pathways Involved in Surviving Extreme ...

<https://journals.asm.org/doi/abs/10.1128/jb.01589-14>

107 Investigating the origins of the mutational signatures in cancer

<https://academic.oup.com/nar/article/53/1/gkae1303/7945387>

113 114 The Effects of Ionizing Radiation on Gut Microbiota - MDPI

<https://www.mdpi.com/1467-3045/45/5/249>

120 121 Frontiers | An Extended Dose–Response Model for Microbial Responses to Ionizing Radiation

<https://www.frontiersin.org/journals/environmental-science/articles/10.3389/fenvs.2017.00006/full>